SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

Beginner

GRAPHS OF TRIGONOMETRIC FUNCTIONS

INVERSE TRIGONOMETRIC FUNCTIONS

Session outline

1. Trigonometric graphs

2. Inverse Trigonometric Functions

Learning objectives

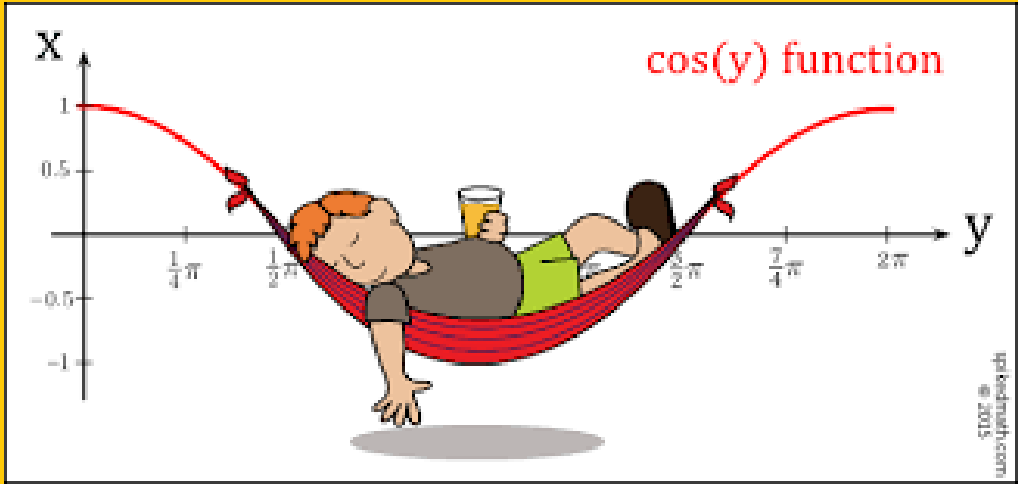
Understand Trigonometric Graphs, including their periodic properties and transformations

Inverse Trigonometric Functions: Understand the concept of inverse trigonometric functions, including how to restrict domains to create one-to-one functions and evaluate these functions.

Simplifying Expressions: Learn to simplify expressions involving trigonometric functions and their inverses, and solve for angles in right triangles

Amplitude and Period: Understand the concepts of amplitude and period, and how they affect the graph of trigonometric functions

1. Trigonometric graphs



Motivation

How do we

model periodic phenomena,

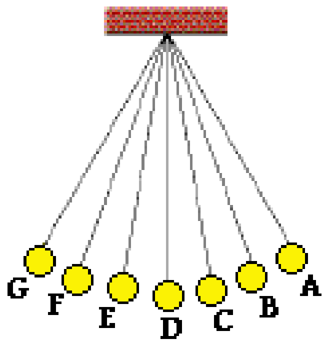
such as the amount of sunlight or variation in

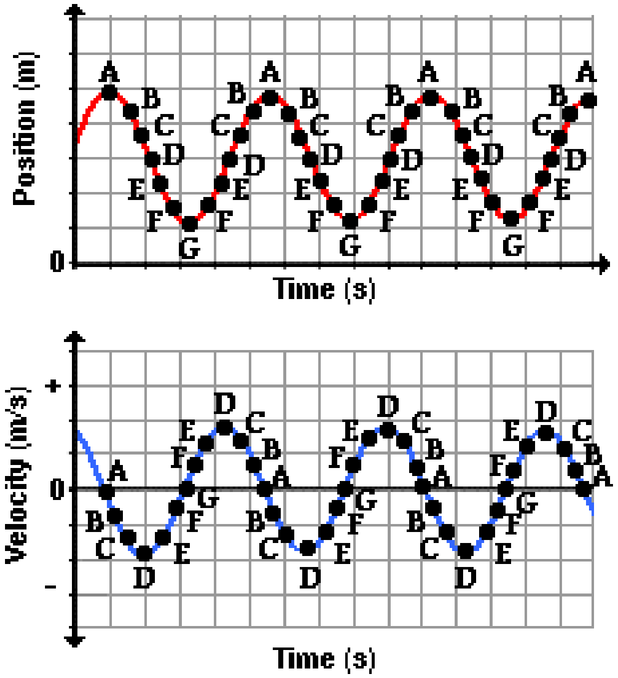
temperature

during the days of the year?

How do we **model harmonic motions** such as

that of pendulums or springs?





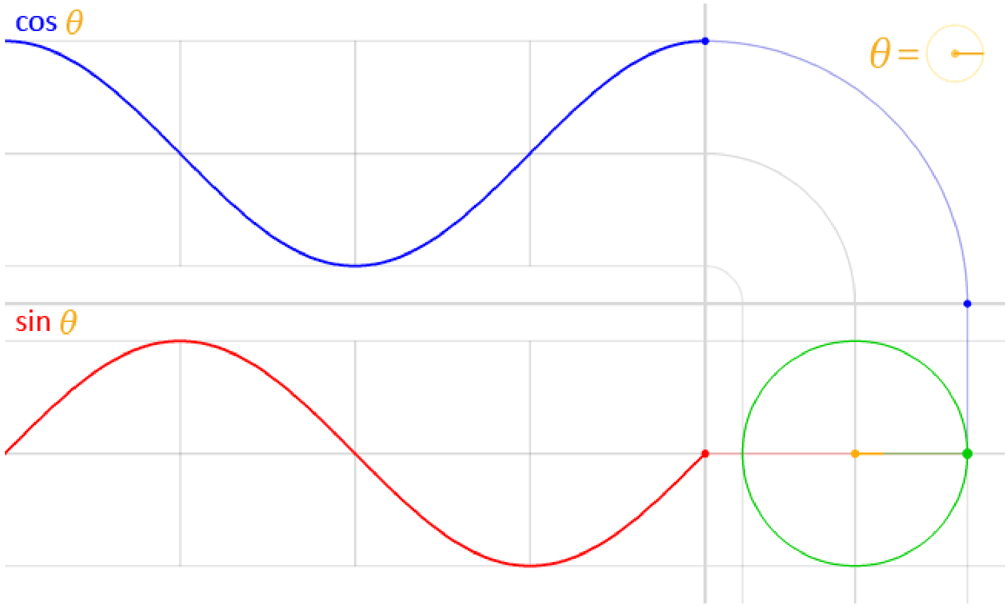
Graphing sine and cosine functions

Periodic properties of Trig functions

PERIODIC PROPERTIES OF SINE AND COSINE

The functions sine and cosine have period 2π:

$$\sin(t + 2\pi) = \sin t \qquad \cos(t + 2\pi) = \cos t$$



PERIODIC:

these functions **repeat their values** in a regular fashion

Why?

circumference of the unit circle is $\mathbf{2\pi}$

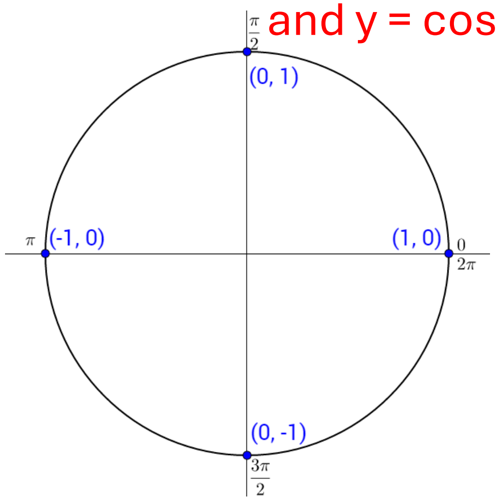
When you rotate a point around the unit circle by an angle of 2π radians (360 degrees), you **complete one full rotation** and **return to the same point** on the circle.

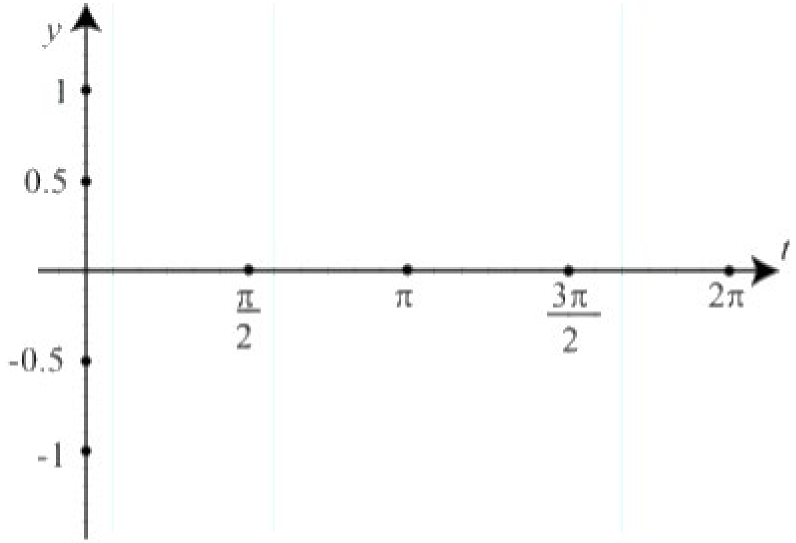
Therefore, the coordinates of the point, and thus the values of the sine and cosine functions, are **the same every 2π radians**.

Graphing activity

Use the values in the given table to plot the points on the graph of y=sin(x) and then draw the graph of y=sin(t) for $0<t<2\pi$.

and y = cos (t)





t	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
sin t	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0
cos t	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1

ACTIVITY 1

Upload your graphs!

1 Graphing activity

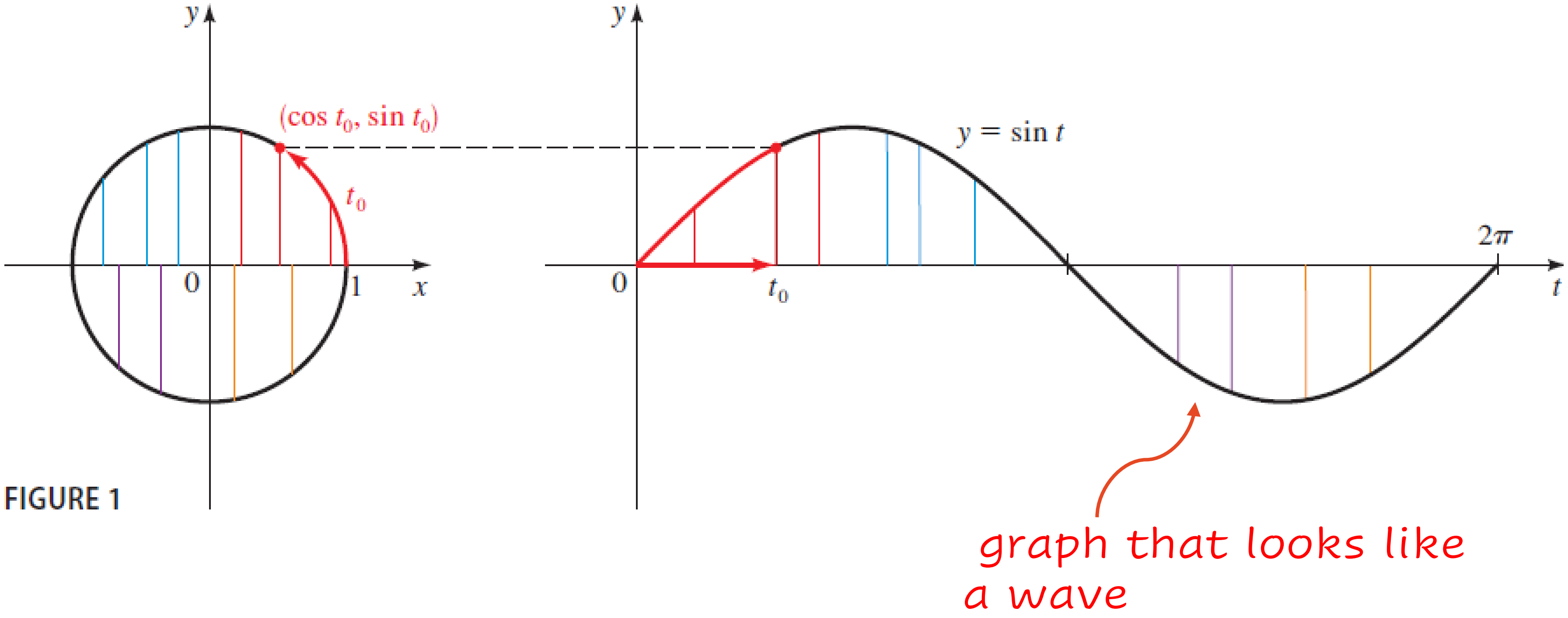


FIGURE 1

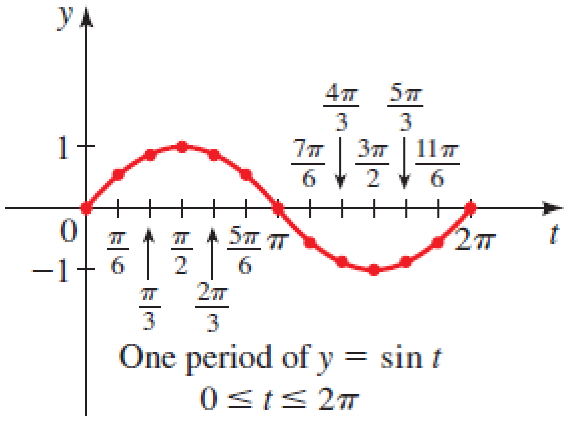
Notice how the **sine values are positive between 0 and π**, which correspond to the values of the sine function in **quadrants I and II** on the unit circle

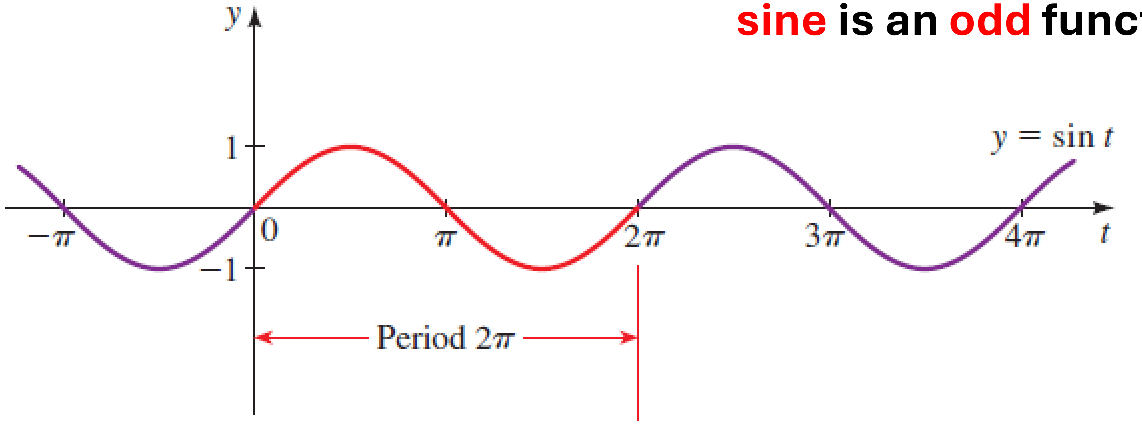
and the **sine values are negative between π and 2π**, which correspond to the values of the sine function in **quadrants III and IV** on the unit circle.

SOLUTION

Graphs of sine and cosine

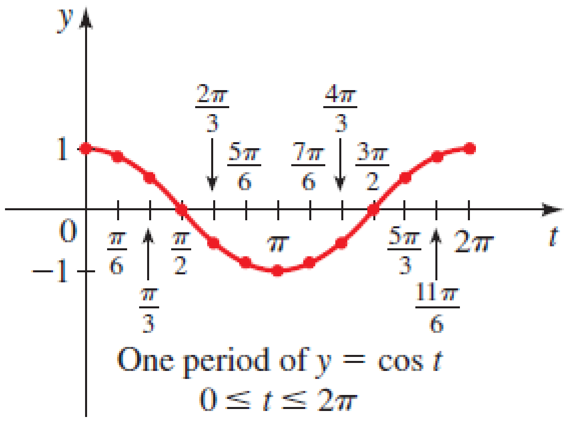
Graph of sin t

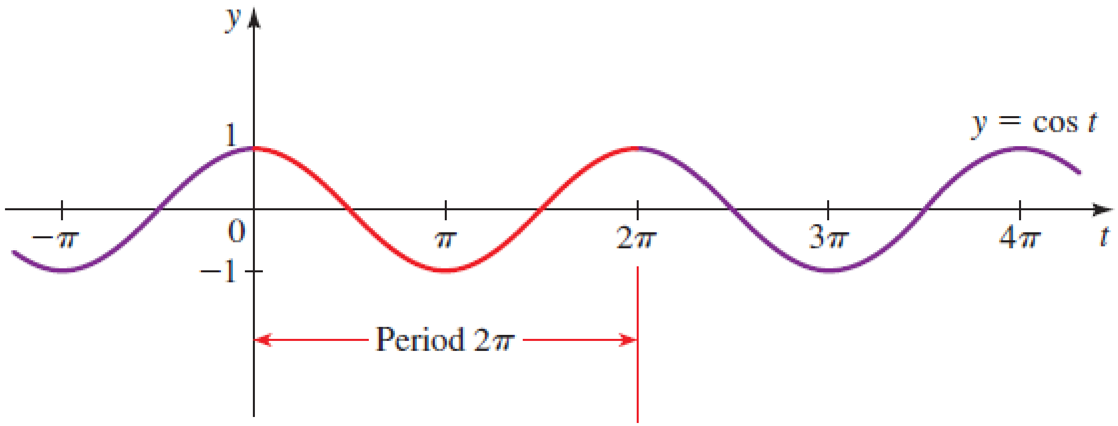




symmetric with respect to the origin
sine is an odd function

Graph of cos t





symmetric with respect to y-axis
cosine is an even function

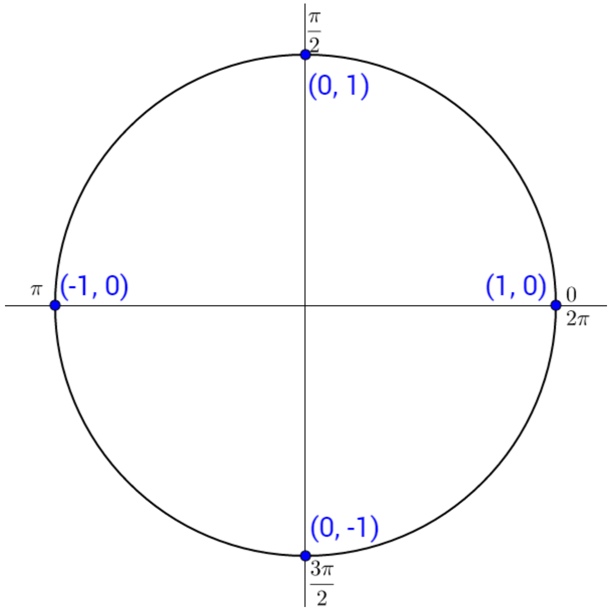
Domain of the trigonometric functions

DOMAINS OF THE TRIGONOMETRIC FUNCTIONS

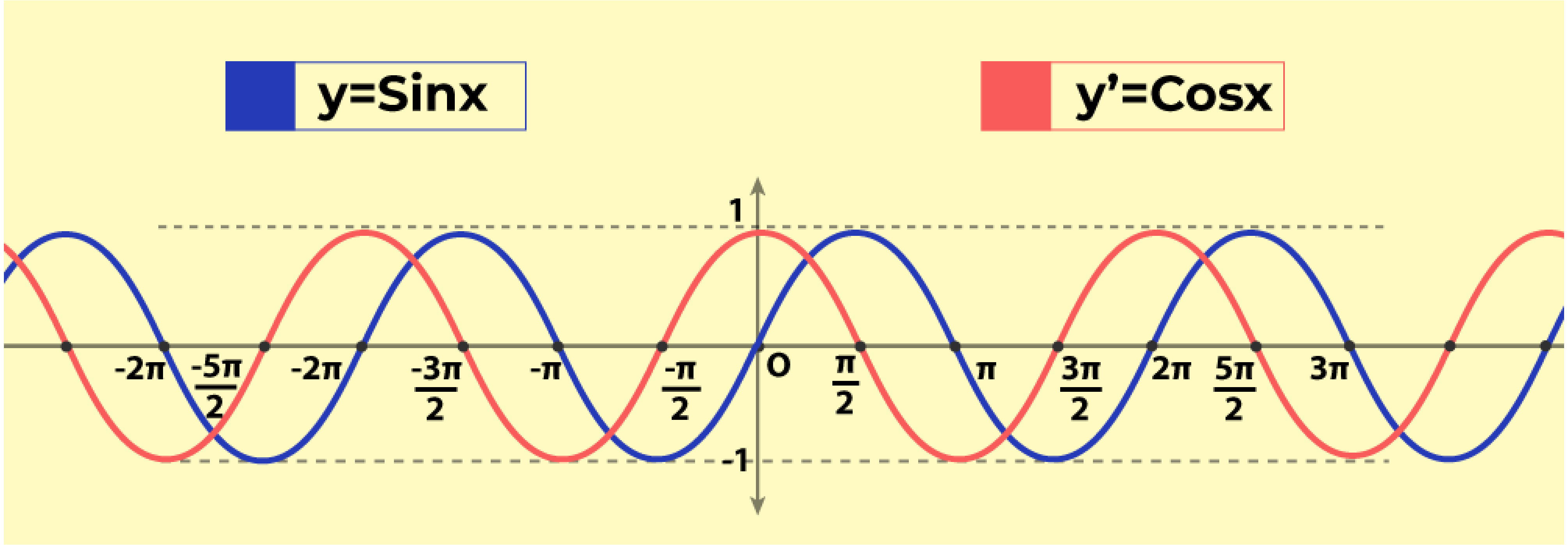
Function	**Domain**
sin, cos	All real numbers
tan, sec	All real numbers other than $\frac{\pi}{2} + n\pi$ for any integer n
cot, csc	All real numbers other than $n\pi$ for any integer, n

Because **we can evaluate the sine and cosine of any real number**, both of these functions are defined for **all real numbers.**

By thinking of the sine and cosine values as coordinates of points on a unit circle, it becomes clear that the **range of both functions must be the interval $[-1,1]$.**



Sine and Cosine function



Graphing tangent function

Graph of tangent

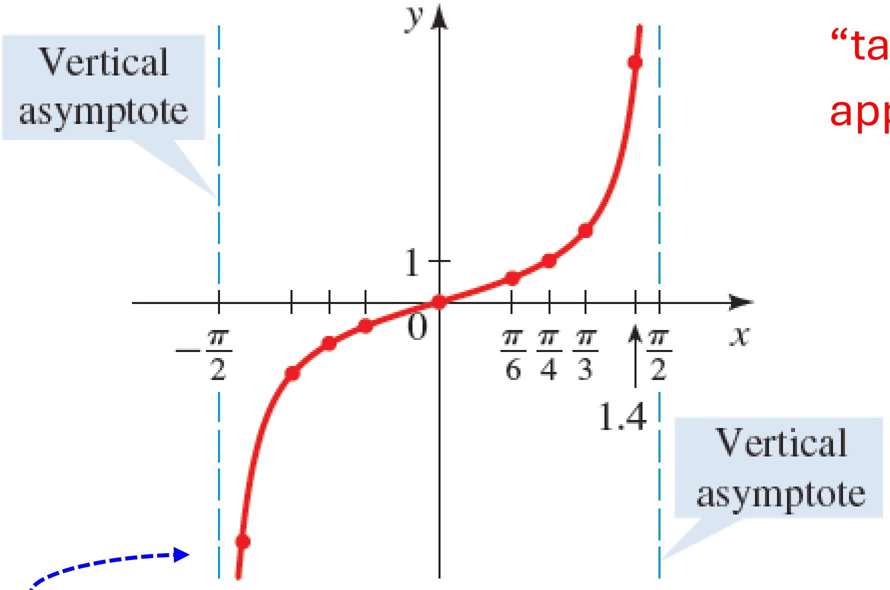
$$tanx = \frac{sinx}{cosx}$$

Tangent function has **vertical asymptotes** where the **cosine function equals zero** (since division by zero is undefined).

As $\boldsymbol{x}$ **approaches** $\frac{\pi}{2}$ from the left, the value of $\boldsymbol{tan\ x}$ **increases** without bound.

$$\tan x \to \infty \qquad \text{as} \qquad x \to \frac{\pi}{2}^-$$

"tan *x* approaches infinity as *x* approaches $\frac{\pi}{2}$ from the left."



As $\boldsymbol{x}$ **approaches** $= -\frac{\pi}{2}$ from the right, the value of $\boldsymbol{tan\ x}$ **decreases** without bound.

$$\tan x \to -\infty \qquad \text{as} \qquad x \to -\frac{\pi}{2}^+$$

"tan *x* approaches negative infinity as *x* approaches $-\frac{\pi}{2}$ from the right."

Since the tangent function has a **period of π**, the pattern of the curve is repeated after every π radians.

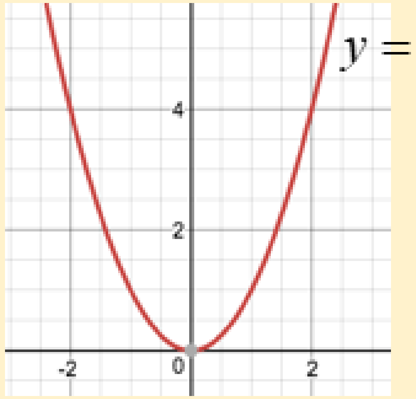
Recall: Even and Odd functions

Even Functions

$$f(-x) = f(x)$$

Function is unchanged when reflected about the y-axis.

Example:



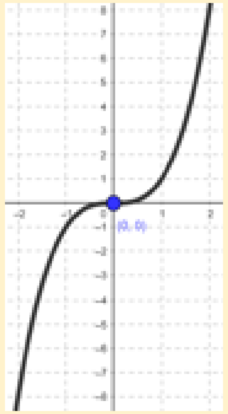
$y = x^2$

Odd Functions

$$f(-x) = -f(x)$$

Function is unchanged when rotated 180° about the origin.

Example:



$y = x^3$

Is tangent an odd or even function?

$$\tan(-x) = \frac{\sin(-x)}{\cos(-x)}$$ Definition of tangent

$$= \frac{-\sin x}{\cos x}$$ Sine is an odd function, cosine is even

$$= -\frac{\sin x}{\cos x}$$ The quotient of an odd and an even function is odd

$$= -\tan x$$ Definition of tangent

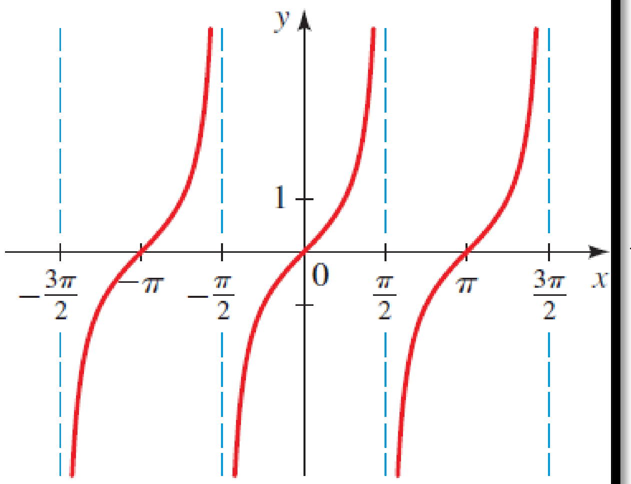
Therefore, tangent is an **odd** function

Graphs of tangent, cotangent, secant, cosecant

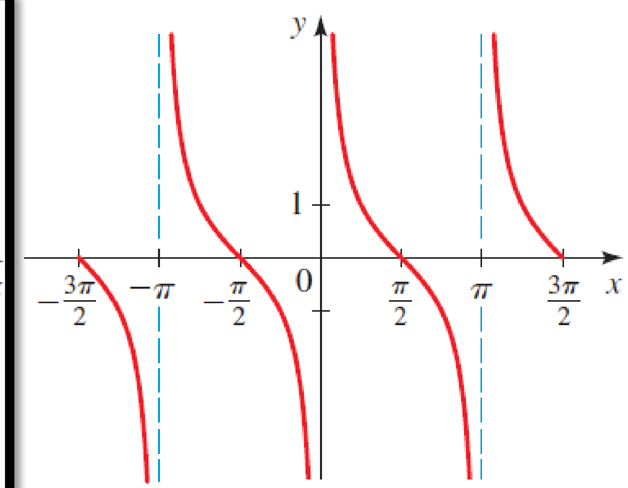
- **Period**: π
- **Asymptotes**: Vertical asymptotes $\frac{\pi}{2} + k\pi$

$$tanx = \frac{sinx}{cosx}$$

- **Symmetry**: Odd



y= tan x



y= cot x

- **Period**: π
- **Asymptotes**: Vertical asymptotes $\mathrm{x} = k\pi$

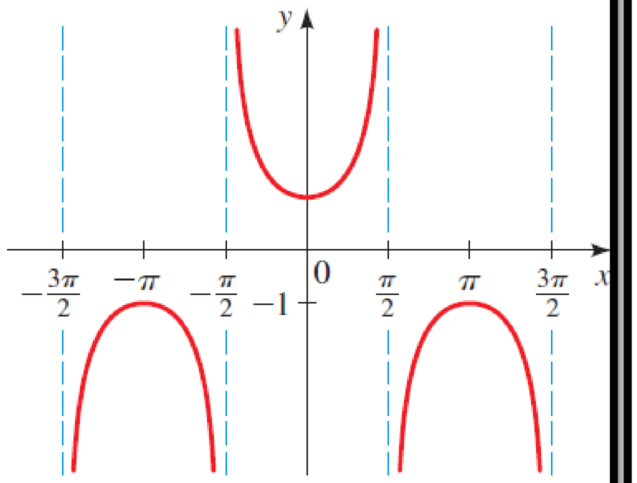
$$\cot x = \frac{\cos x}{\sin x}$$

- **Symmetry**: Odd

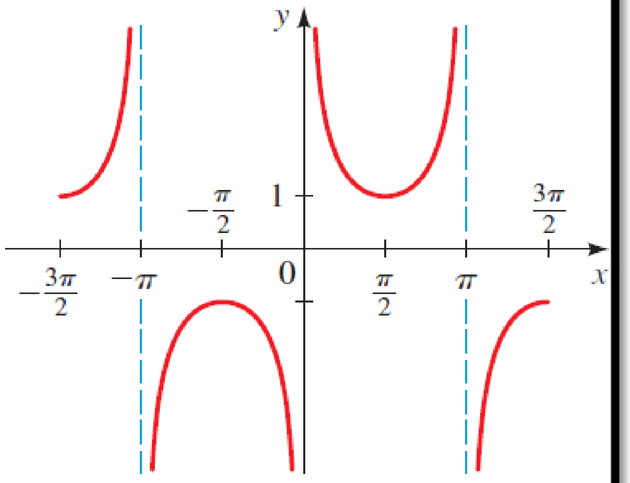
- **Period**: 2π
- **Asymptotes**: Vertical asymptotes $\frac{\pi}{2} + k\pi$

$$secx = \frac{1}{\cos x}$$

- **Symmetry**: Even



y= sec x



y= csc x

- **Period**: 2π
- **Asymptotes**: Vertical asymptotes $x = k\pi$

$$cscx = \frac{1}{sinx}$$

- **Symmetry**: Odd

Transformations of graphs

Transformation rules

Function Notation	Type of Transformation
$f(x) + d$	Vertical translation up d units
$f(x) - d$	Vertical translation down d units
$f(x + c)$	Horizontal translation left c units
$f(x - c)$	Horizontal translation right c units
$-f(x)$	Reflection over x-axis
$f(-x)$	Reflection over y-axis
$af(x)$	Vertical stretch for $\|a\| > 1$
	Vertical compression for $0 < \|a\| < 1$
$f(bx)$	Horizontal compression for $\|b\| > 1$
	Horizontal stretch for $0 < \|b\| < 1$

REVIEW

Sketch the graph of the following Trig functions

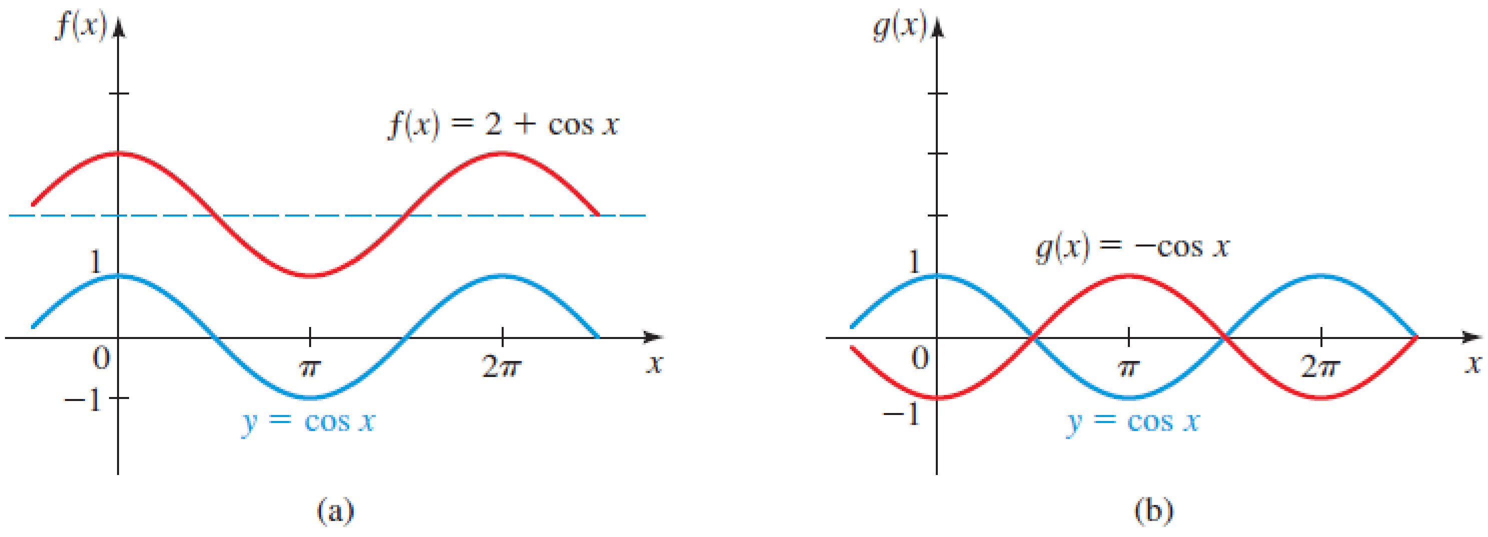
Sketch the graph of each function.

(a) $f(x) = 2 + \cos x$ **(b)** $g(x) = -\cos x$

SOLUTION

(a) The graph of $y = 2 + \cos x$ is the same as the graph of $y = \cos x$, but shifted up 2 units (see Figure 4(a)).

(b) The graph of $y = -\cos x$ in Figure 4(b) is the reflection of the graph of $y = \cos x$ in the x-axis.



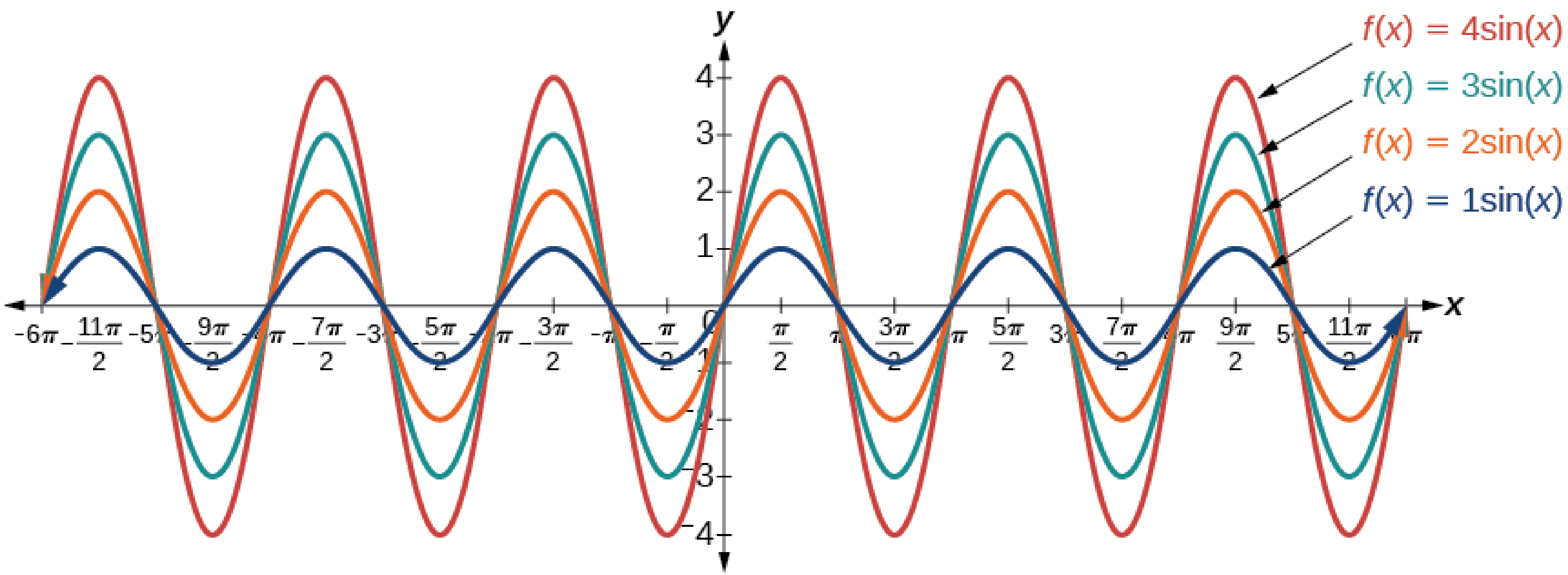
EXAMPLE

Understanding Amplitude

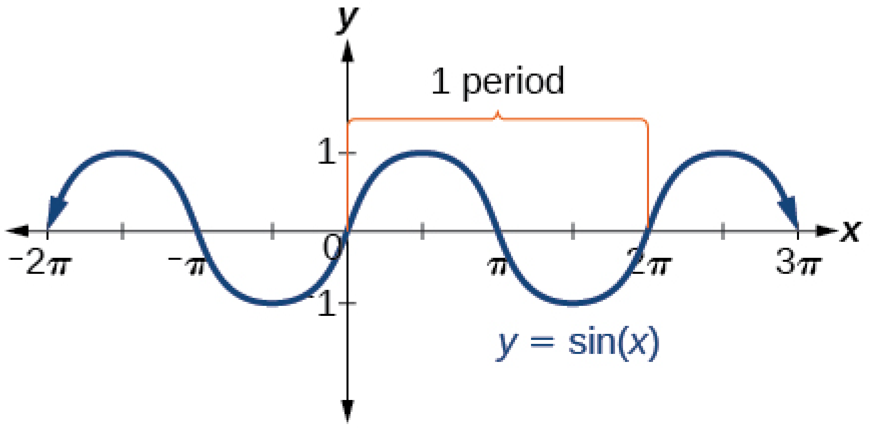
By multiplying a graph by a value, ***a***, the graph is **stretched vertically by a**.

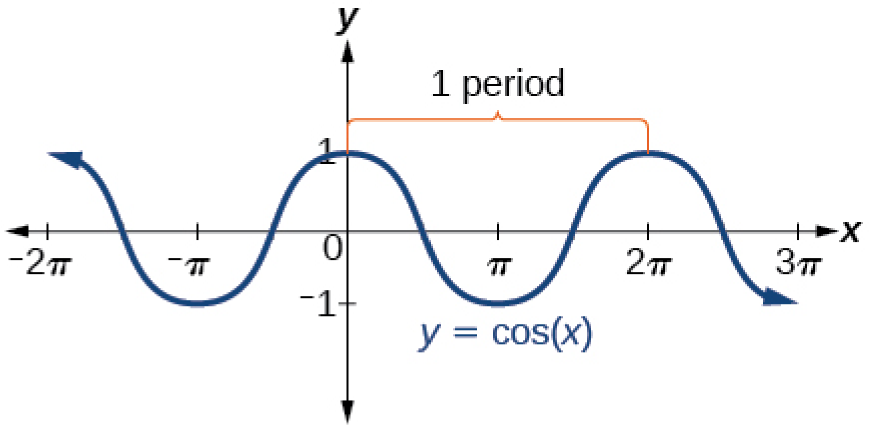
This is because the function equals y, so if we multiply the function by a, then we multiply y by a.

The original height of the sine or cosine graph is 1, the transformed height is a. This height is called the **amplitude**



Understanding period





The **distance on the x-axis of one full cycle** of the graph is **period**.

The period of the basic sine or cosine graph is 2π, but if the x is multiplied by a number, $\boldsymbol{b}$, the graph is **horizontally shrunk by the factor, $\boldsymbol{b}$.**

Amplitude and Period: formal definitions

SINE AND COSINE CURVES

The sine and cosine curves

$$y = a \sin kx \qquad \text{and} \qquad y = a \cos kx \qquad (k > 0)$$

have **amplitude** $|a|$ and **period** $2\pi/k$.

An appropriate interval on which to graph one complete period is $[0, 2\pi/k]$.

Amplitude and Period

Find the amplitude and period of each function, and sketch its graph.

(a) $y = 4\cos 3x$ **(b)** $y = -2\sin \frac{1}{2}x$

SOLUTION

(a) We get the amplitude and period from the form of the function as follows.

$$\text{amplitude} = |a| = 4$$

$$y = 4\cos 3x$$

$$\text{period} = \frac{2\pi}{k} = \frac{2\pi}{3}$$

The amplitude is 4, and the period is $2\pi/3$. The graph is shown in Figure 10.

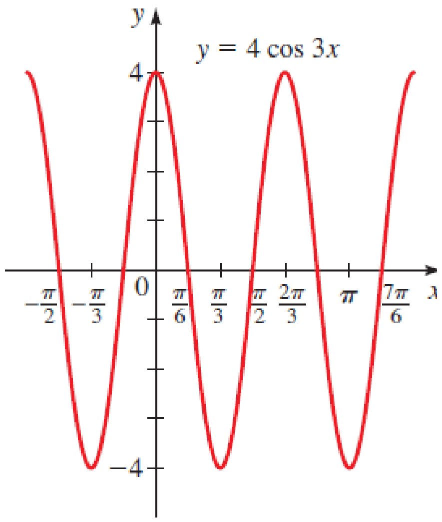


FIGURE 10

(b) For $y = -2\sin \frac{1}{2}x$,

$$\text{amplitude} = |a| = |-2| = 2$$

$$\text{period} = \frac{2\pi}{\frac{1}{2}} = 4\pi$$

The graph is shown in Figure 11.

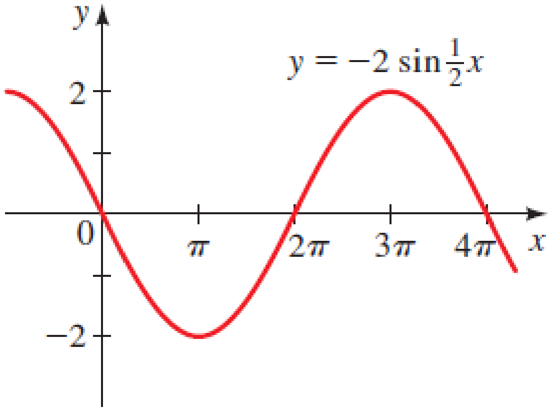


FIGURE 11

EXAMPLE

Graphing sine and cosine functions

(a) $g(x) = 4 - 2\sin x$

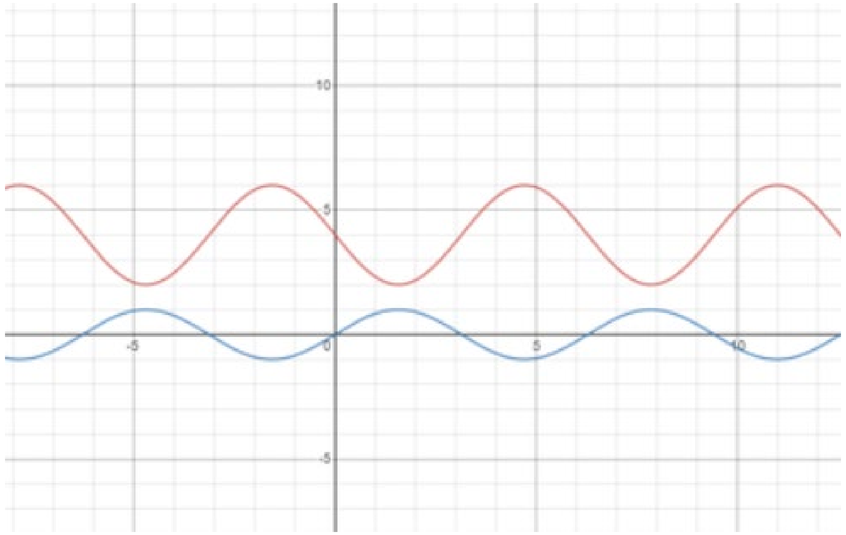
(b) $h(x) = |\sin x|$

ACTIVITY 2

2 Graphing sine and cosine functions

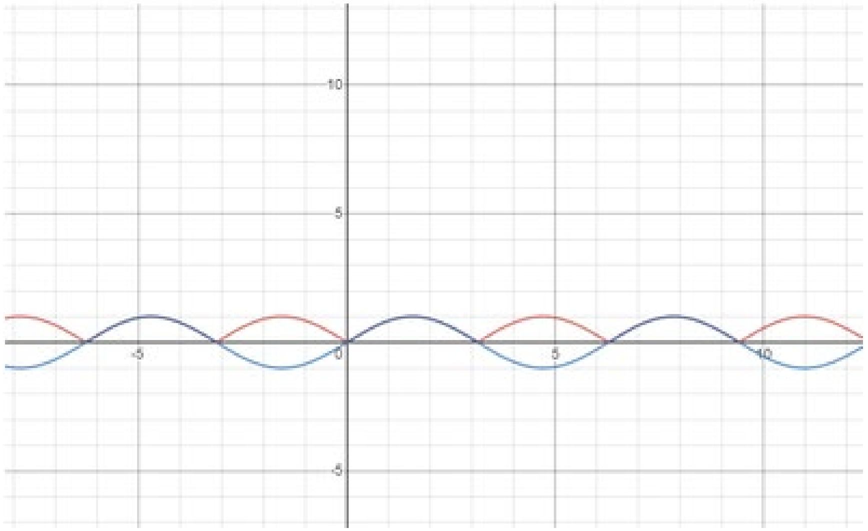
(a) $g(x) = 4 - 2\sin x$

The blue graph represents the graph of $sinx$
The red graph represents the graph of the given function, which is vertically stretched by a factor of 2, reflected about the x-axis, and translated of 4 units up from the blue graph.



(b) $h(x) = |\sin x|$

The blue graph represents the graph of $sinx$
The red graph represents the graph of the given function which is only positive values, thus there is no part of the graph that extends below the x-axis due to the absolute value bars



SOLUTION

Amplitude and period

Find the amplitude and period of the function, and sketch its graph.

(a) $y = 4\sin(-2x)$

(b) $y = -3\sin \pi x$

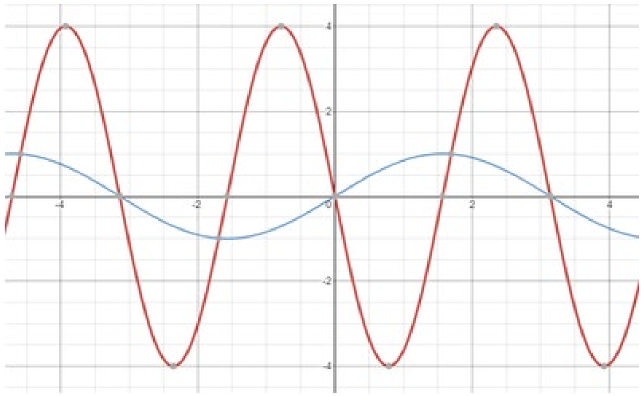
ACTIVITY 3

3 Amplitude and period

(a) $y = 4\sin(-2x)$

Given $y = 4\sin(-2x)$
The Amplitude is $|4| = 4$
The Period is $\frac{2\pi}{|-2|} = \pi$

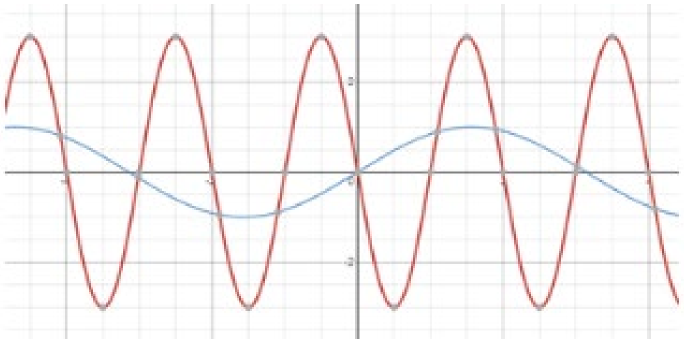
The transformations are as follows
The blue graph represents the graph of $\sin x$
The red graph represents the graph of the given function that is reflected about the y-axis, compressed horizontally by a factor of 0.5 (calculated by $\frac{1}{c}$) and stretched vertically by a factor of 4



(b) $y = -3\sin \pi x$

Given $y = -3\sin(\pi x)$
The Amplitude is $|-3| = 3$
The Period is $\frac{2\pi}{|\pi|} = 2$

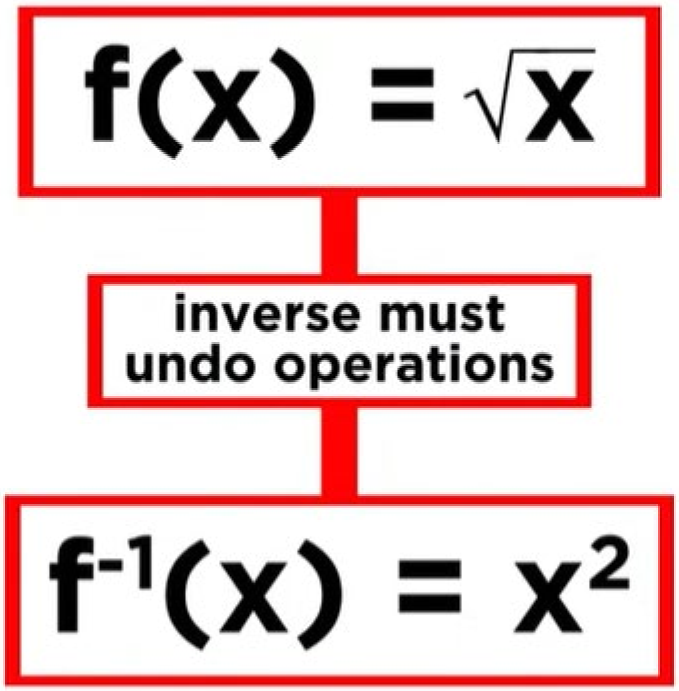
The transformations are as follows
The blue graph represents the graph of $\sin x$
The red graph represents the graph of the given function that is reflected about the x-axis, compressed horizontally by a factor of π and stretched vertically by a factor of 3



SOLUTION

2. Inverse Trigonometric Functions

Recall what is an inverse function?



Do Trigonometric functions have inverses?

Trigonometric functions are all **periodic functions** .

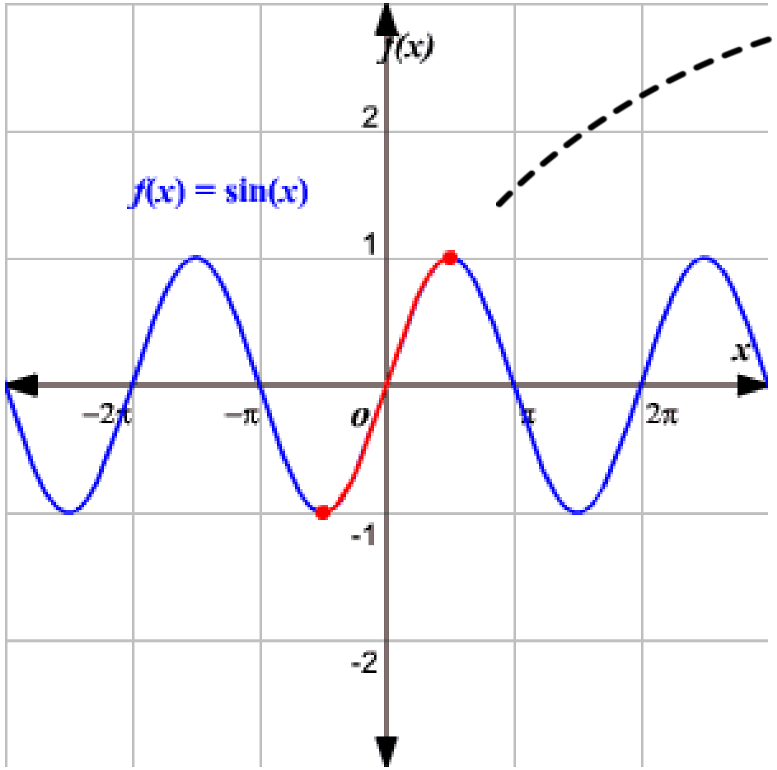
Thus the graphs of none of them pass the **Horizontal Line Test** and so are not **One to One.**

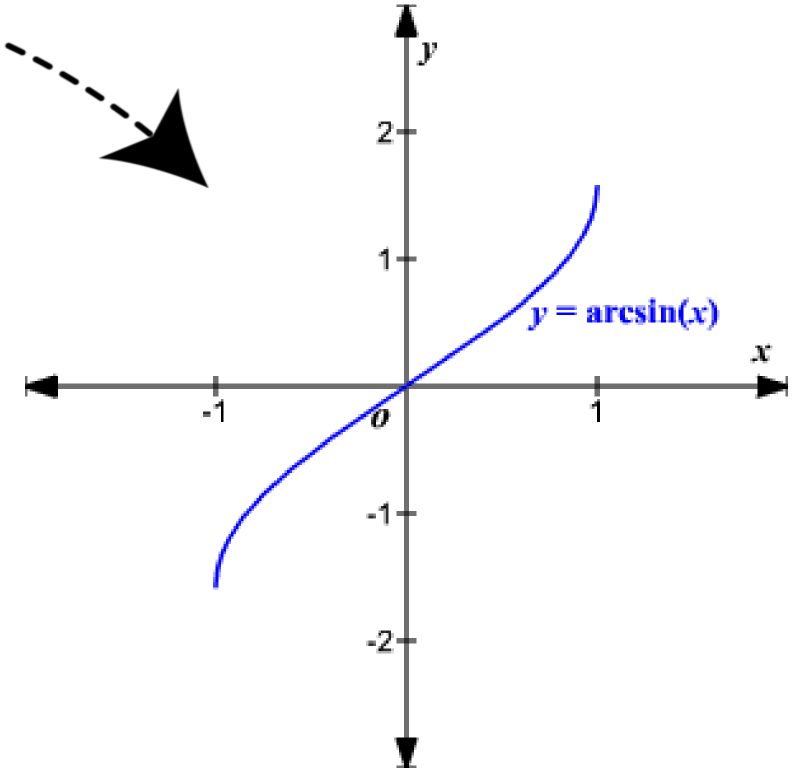
This means none of them have an inverse **unless the domain of each is restricted** to make each of them **One to one.**

functions must pass the horizontal line test to have an inverse function

Restricting domain of the Sine function to create one to one function

Take this piece and reflect it along y=x





If we restrict the domain of $f(x)$=sin(x) to **[$-\pi/2,\pi/2$]** we have made the function one to one.

The range is [−1,1] .

Understanding the sine inverse function

$$y = \sin x$$

$$x = \sin y$$

swap variables and solve for y

$$\sin^{-1}(x) = \sin^{-1}(\sin y)$$

we must take the inverse sine of both sides

$$y = \sin^{-1} x$$

This is read as "y is the inverse of sine x"
and means y is the **real number angle** whose sine value is x .

sine		inverse sine	
y	$= \sin x$	y	$= \sin^{-1} x$
↑ sine value	↑ angle	↑ corresponding angle	↑ valid sine value

Note: The '-1' is not an exponent!

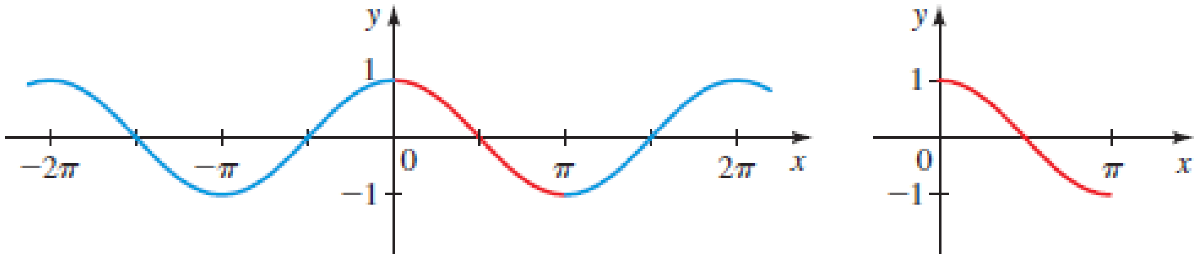
$$\boxed{y = \sin^{-1} x}$$

$$\sin^{-1} x \neq \frac{1}{\sin x}$$

To avoid this confusion, some books use the notation y=**arcsin**(x) to indicate the inverse

$$(\sin x)^{-1} = \frac{1}{\sin x} = \csc x$$

Restricting domain of the Cosine function



$y = \cos x$ $y = \cos x,\ 0 \leq x \leq \pi$

If the domain of the cosine function is restricted to the interval $[0, \pi]$, the resulting function is **one-to-one** and so has an inverse.

DEFINITION OF THE INVERSE COSINE FUNCTION

The **inverse cosine function** is the function $\cos^{-1}$ with domain $[-1, 1]$ and range $[0, \pi]$ defined by

$$\cos^{-1} x = y \quad \Leftrightarrow \quad \cos y = x$$

The inverse cosine function is also called **arccosine**, denoted by **arccos**.

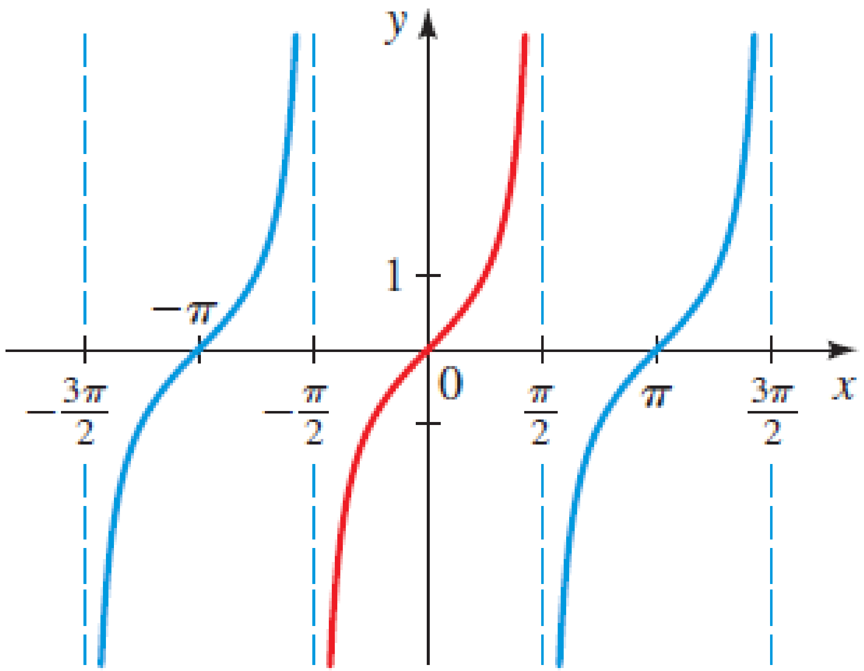
Restricting domain of the Tangent function

DEFINITION OF THE INVERSE TANGENT FUNCTION

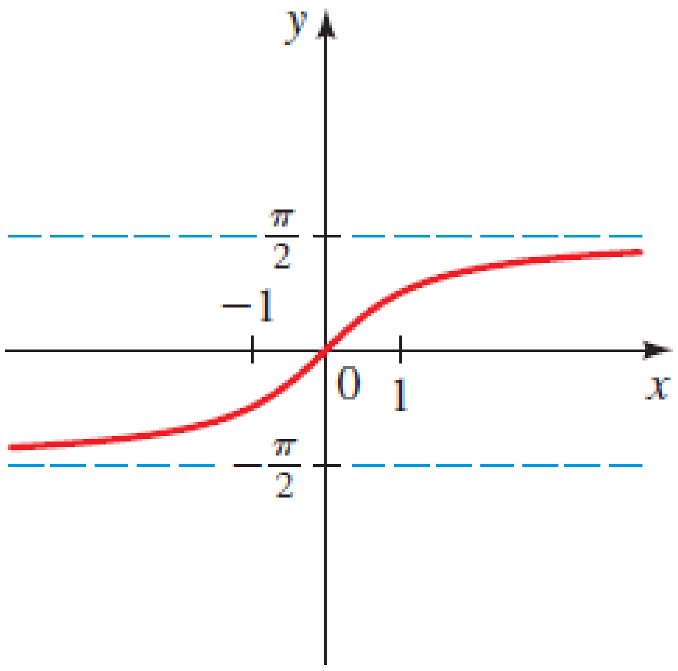
The **inverse tangent function** is the function $\tan^{-1}$ with domain $\mathbb{R}$ and range $(-\pi/2, \pi/2)$ defined by

$$\tan^{-1} x = y \quad \Leftrightarrow \quad \tan y = x$$

The inverse tangent function is also called **arctangent**, denoted by **arctan**.



$y = \tan x,\ -\frac{\pi}{2} < x < \frac{\pi}{2}$



$y = \tan^{-1} x$

Cancellation properties of inverse functions

Recall the inverse function property we saw previously...

$$f^{-1}(f(x)) = x$$
$$f(f^{-1}(x)) = x$$

applying a function and its inverse
in succession
will return
the original value

$$\sin(\sin^{-1} x) = x \quad \text{for} \quad -1 \le x \le 1$$

Domain of arcsine

$$\sin^{-1}(\sin x) = x \quad \text{for} \quad -\frac{\pi}{2} \le x \le \frac{\pi}{2}$$

Domain of sine

$$\cos(\cos^{-1} x) = x \quad \text{for} \quad -1 \le x \le 1$$

Domain of arccosine

$$\cos^{-1}(\cos x) = x \quad \text{for} \quad 0 \le x \le \pi$$

Domain of cosine

$$\tan(\tan^{-1} x) = x \quad \text{for} \quad x \in \mathbb{R}$$

Domain of arctangent

$$\tan^{-1}(\tan x) = x \quad \text{for} \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

Domain of tangent

The Inverse Sine, Inverse Cosine, and Inverse Tangent Functions

THE INVERSE SINE, INVERSE COSINE, AND INVERSE TANGENT FUNCTIONS

The sine, cosine, and tangent functions on the restricted domains $[-\pi/2, \pi/2]$, $[0, \pi]$, and $(-\pi/2, \pi/2)$, respectively, are one-to one and so have inverses. The inverse functions have domain and range as follows.

Function	Domain	Range
$\sin^{-1}$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

The functions $\sin^{-1}$, $\cos^{-1}$, and $\tan^{-1}$ are sometimes called **arcsine**, **arccosine**, and **arctangent**, respectively.

Evaluating Inverse Trigonometric Functions

Find the exact value.

(a) $\sin^{-1} \dfrac{\sqrt{3}}{2}$ **(b)** $\cos^{-1}(-\frac{1}{2})$ **(c)** $\tan^{-1} 1$

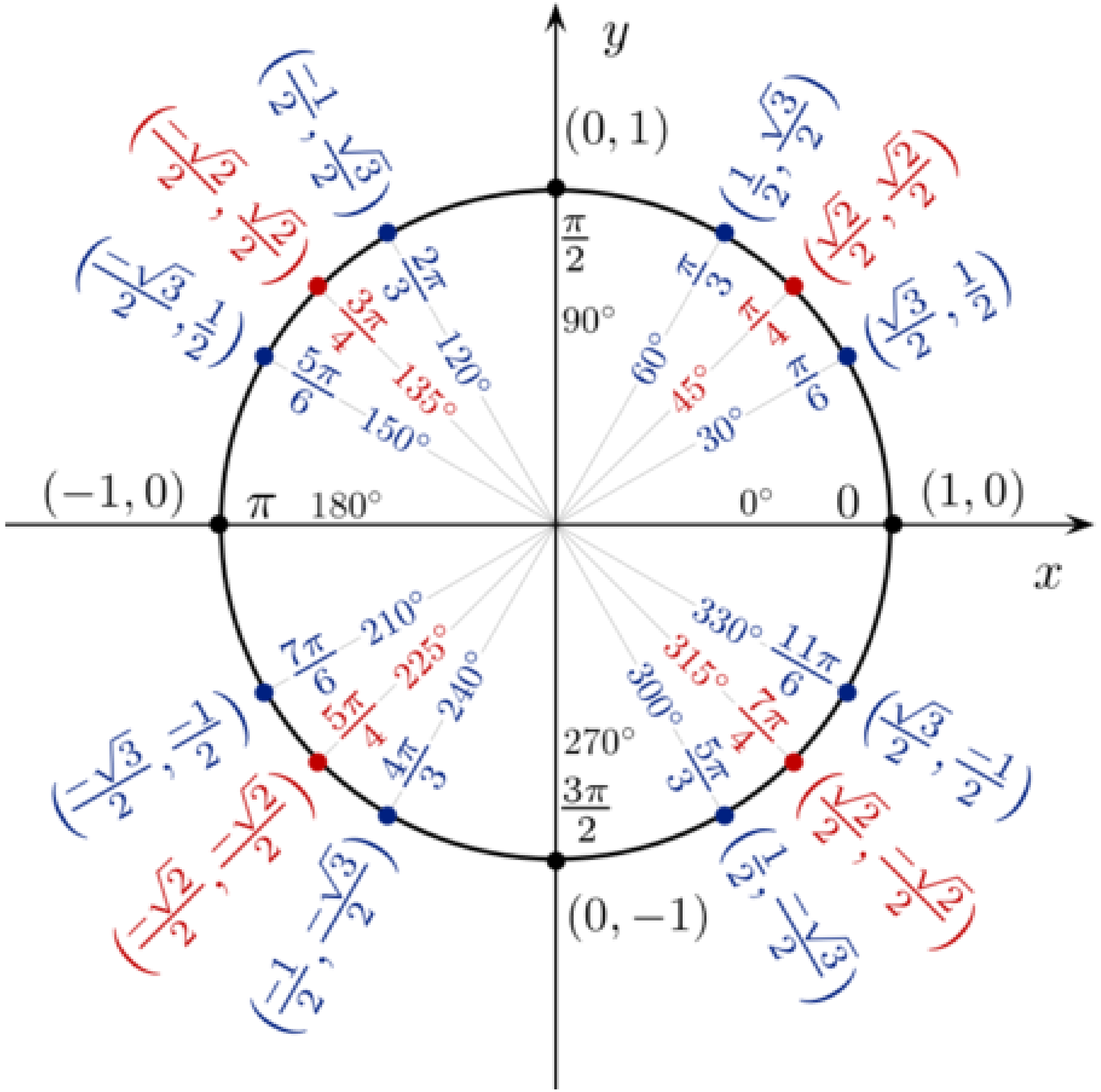
SOLUTION

(a) The angle in the interval $[-\pi/2, \pi/2]$ whose sine is $\sqrt{3}/2$ is $\pi/3$. Thus $\sin^{-1}(\sqrt{3}/2) = \pi/3$.

(b) The angle in the interval $[0, \pi]$ whose cosine is $-\frac{1}{2}$ is $2\pi/3$. Thus $\cos^{-1}(-\frac{1}{2}) = 2\pi/3$.

(c) The angle in the interval $(-\pi/2, \pi/2)$ whose tangent is 1 is $\pi/4$. Thus $\tan^{-1} 1 = \pi/4$.

EXAMPLE



y
x
$(0, 1)$
$(1, 0)$
$(-1, 0)$
$(0, -1)$
$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
$\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
$\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
$\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
$\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
$\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
$\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
$\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
$\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
$\frac{\pi}{2}$
$90°$
$\frac{\pi}{3}$
$60°$
$\frac{\pi}{4}$
$45°$
$\frac{\pi}{6}$
$30°$
$0°$
0
$\frac{2\pi}{3}$
$120°$
$\frac{3\pi}{4}$
$135°$
$\frac{5\pi}{6}$
$150°$
π
$180°$
$\frac{7\pi}{6}$
$210°$
$\frac{5\pi}{4}$
$225°$
$\frac{4\pi}{3}$
$240°$
$270°$
$\frac{3\pi}{2}$
$\frac{5\pi}{3}$
$300°$
$\frac{7\pi}{4}$
$315°$
$\frac{11\pi}{6}$
$330°$

Evaluating inverse trigonometric functions

(a) $\cos^{-1}\left(-\frac{1}{2}\right)$ **(b)** $\sin^{-1}\left(-\frac{\sqrt{2}}{2}\right)$ **(c)** $\tan^{-1} 1$

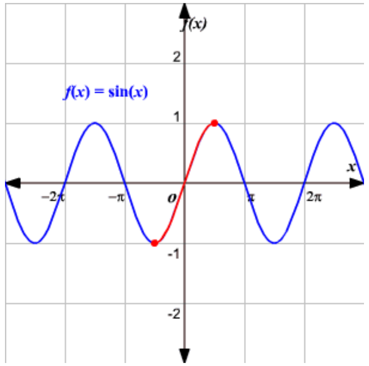
cosine of what angle will
give us $-\frac{1}{2}$?

sine of what angle will
give us $-\frac{\sqrt{2}}{2}$?

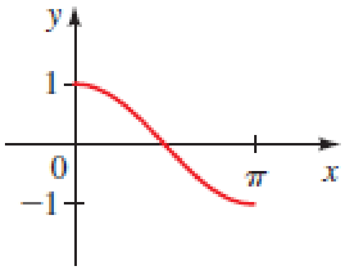
ACTIVITY 4

4 Evaluating inverse trigonometric functions

a) First, we look at the interval of the inverse of cosine which is $0 < x < \pi$. Then, we must find what angle will give a value of $-1/2$ in that interval. The answer is $\frac{2\pi}{3}$

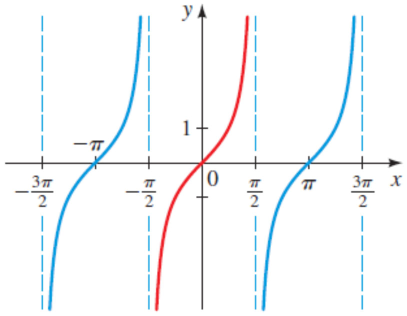


b) First, we look at the interval of the inverse of sine which is $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Then, we must find what angle will give a value of $-\sqrt{2}/2$ in that interval. The answer is $-\frac{\pi}{4}$



$y = \cos x,\ 0 \leq x \leq \pi$

c) First, we look at the interval of the inverse of tangent which is $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Then, we must find what angle will give a slope of 1 in that interval. The answer is $\frac{\pi}{4}$



$y = \tan x,\ -\frac{\pi}{2} < x < \frac{\pi}{2}$

SOLUTION

Simplifying expressions with Trig functions

(a) $\cos\left(\cos^{-1}\frac{2}{3}\right)$

(b) $\sin\left(\sin^{-1} 5\right)$

(c) $\tan\left(\tan^{-1}\frac{3}{2}\right)$

Is there a property that we can use?

What do you need to be careful about when using the property?

ACTIVITY 5

5a Simplifying expressions with Trig functions

(a) $\cos\left(\cos^{-1}\frac{2}{3}\right)$

$$\cos(\cos^{-1}x) = x \quad \text{for} \quad -1 \le x \le 1$$
$$\cos^{-1}(\cos x) = x \quad \text{for} \quad 0 \le x \le \pi$$

Cancellation properties
of inverse functions

It is clear the answer will either be the number in the parentheses or it will be undefined.

If the value that we take to be inverted is greater than 1 or less than -1 for a sine or cosine function, the value overall isn't defined.

Here the value is in the (-1,1) range so the answer is $\frac{2}{3}$

SOLUTION

5b Simplifying expressions with Trig functions

(b) $\sin(\sin^{-1} 5)$

$$\sin(\sin^{-1} x) = x \quad \text{for} \quad -1 \leq x \leq 1$$
$$\sin^{-1}(\sin x) = x \quad \text{for} \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

Cancellation properties of inverse functions

UNDEFINED

Whenever the value that we take to be inverted is greater than 1 or less than -1 for a sine or cosine function, the value overall isn't defined.

SOLUTION

Solving for Angles in a Right triangle

Find the angle θ in the triangle shown in Figure 2.

SOLUTION Since θ is the angle opposite the side of length 10 and the hypotenuse has length 50, we have

$$\sin\theta = \frac{10}{50} = \frac{1}{5} \qquad \sin\theta = \frac{\text{opp}}{\text{hyp}}$$

Now we can use $\sin^{-1}$ to find θ.

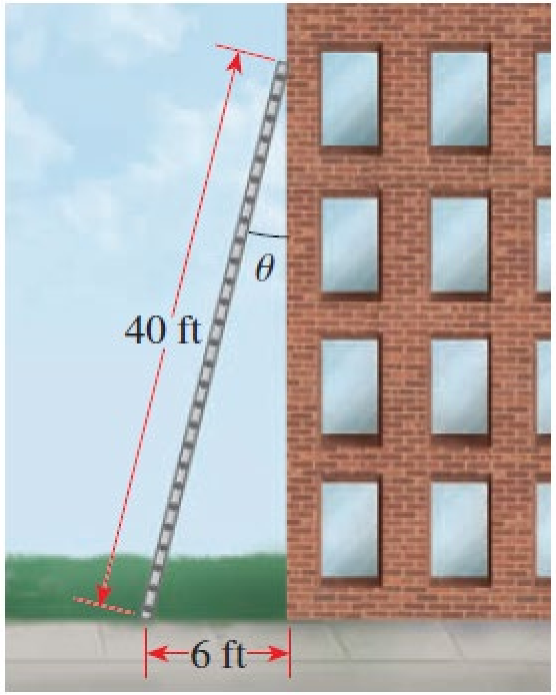
$$\theta = \sin^{-1}\tfrac{1}{5} \qquad \text{Definition of } \sin^{-1}$$

$$\theta \approx 11.5^\circ \qquad \text{Calculator (in degree mode)}$$

EXAMPLE

Solving for Angles in a Right triangle

A 40-ft ladder leans against a building. If the base of the ladder is 6 ft from the base of the building, what is the angle formed by the ladder and the building?



SOLUTION First we sketch a diagram as in Figure 3. If θ is the angle between the ladder and the building, then

$$\sin\theta = \frac{6}{40} = 0.15 \qquad \sin\theta = \frac{\text{opp}}{\text{hyp}}$$

Now we use $\sin^{-1}$ to find θ.

$$\theta = \sin^{-1}(0.15) \qquad \text{Definition of } \sin^{-1}$$

$$\theta \approx 8.6° \qquad \text{Calculator (in degree mode)}$$

EXAMPLE

Evaluating Expressions Involving Inverse Trigonometric Functions

Find $\cos\left(\sin^{-1} \frac{3}{5}\right)$.

SOLUTION 1 Let $\theta = \sin^{-1} \frac{3}{5}$. Then θ is the number in the interval $[-\pi/2, \pi/2]$ whose sine is $\frac{3}{5}$. Let's interpret θ as an angle and draw a right triangle with θ as one of its acute angles, with opposite side 3 and hypotenuse 5 (see Figure 6). The remaining leg of the triangle is found by the Pythagorean Theorem to be 4. From the figure we get

$$\cos\left(\sin^{-1} \tfrac{3}{5}\right) = \cos\theta \qquad \theta = \sin^{-1} \tfrac{3}{5}$$

$$= \frac{4}{5} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}}$$

So $\cos\left(\sin^{-1} \frac{3}{5}\right) = \frac{4}{5}$.

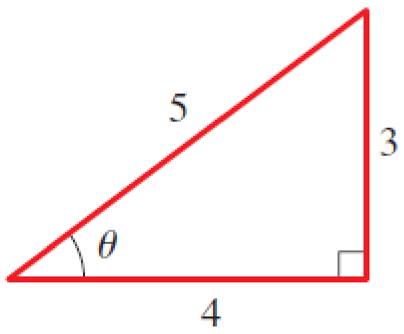


FIGURE 6

$\cos\theta = \dfrac{4}{5}$

EXAMPLE

Composing Trigonometric Functions and Their Inverses

Write $\sin(\cos^{-1}x)$ and $\tan(\cos^{-1}x)$ as algebraic expressions in x for $-1 \le x \le 1$.

SOLUTION 1 Let $\theta = \cos^{-1}x$; then $\cos\theta = x$. In Figure 7 we sketch a right triangle with an acute angle θ, adjacent side x, and hypotenuse 1. By the Pythagorean Theorem the remaining leg is $\sqrt{1 - x^2}$. From the figure we have

$$\sin(\cos^{-1}x) = \sin\theta = \sqrt{1 - x^2} \qquad \text{and} \qquad \tan(\cos^{-1}x) = \tan\theta = \frac{\sqrt{1 - x^2}}{x}$$

SOLUTION 2 Let $u = \cos^{-1}x$. We need to find $\sin u$ and $\tan u$ in terms of x. As in Example 7 the idea here is to write sine and tangent in terms of cosine. Note that $0 \le u \le \pi$ because $u = \cos^{-1}x$. We have

$$\sin u = \pm\sqrt{1 - \cos^2 u} \qquad \text{and} \qquad \tan u = \frac{\sin u}{\cos u} = \frac{\pm\sqrt{1 - \cos^2 u}}{\cos u}$$

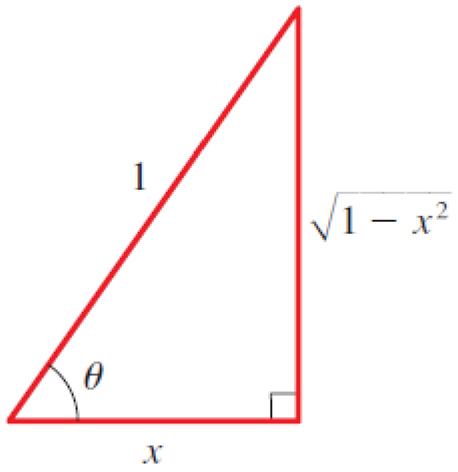


FIGURE 7

$$\cos\theta = \frac{x}{1} = x$$

EXAMPLE

Value of an expression

(a) $\cos\left(\tan^{-1}\frac{4}{3}\right)$

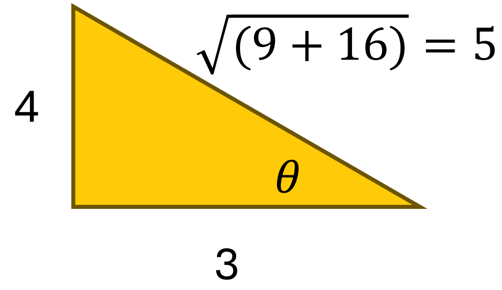
(b) $\csc\left(\cos^{-1}\frac{7}{25}\right)$

ACTIVITY 6

6 Value of an expression

(a) $\cos\left(\tan^{-1}\frac{4}{3}\right)$

Let $\theta = \tan^{-1}\left(\frac{4}{3}\right)$



$\tan\theta = \frac{4}{3}$

$cos\theta = \frac{3}{5}$

$\cos\left(tan^{-1}\frac{4}{3}\right) = \frac{3}{5}$

SOLUTION

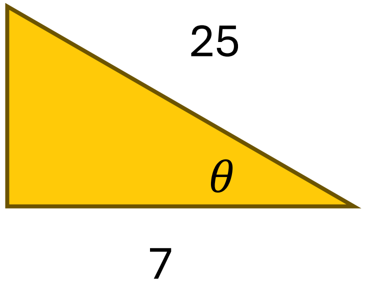
6 Value of an expression

(b) $\csc\left(\cos^{-1}\frac{7}{25}\right)$

Let $\theta = \cos^{-1}\left(\frac{7}{25}\right)$

$$\cos\theta = \frac{7}{25}$$

$$\sqrt{625-49} = 24$$



$$\csc\left(\cos^{-1}\frac{7}{25}\right) = csc\theta = \frac{hypotenuse}{perpendicular} = \frac{25}{24}$$

SOLUTION

Rewrite expressions

Rewrite the expression as an **algebraic expression** in x.

(a) $\sin(\tan^{-1} x)$

(b) $\tan(\sin^{-1} x)$

Assume x is positive

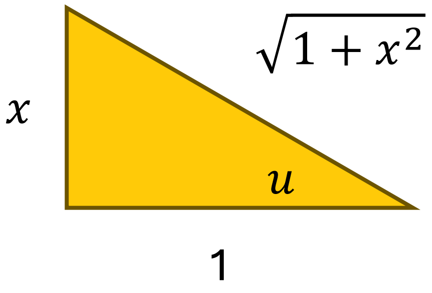
ACTIVITY 7

7 Rewrite expressions

(a) $\sin(\tan^{-1} x)$

$\boxed{\text{Assuming } \tan^{-1} x = u\,,}$

$\tan u = x$, $[-\frac{\pi}{2} \le u \le \frac{\pi}{2}]$



Considering right triangle ABC in which, $\tan u = x$

i.e. in right triangle ABC, $\frac{opp}{adj} = \frac{x}{1}$

From Pythagorean Theorem we know

$hypo = \sqrt{adj^2 + opp^2}$

Therefore-

$\sin u = \frac{opp}{hypo}$ i.e. $\sin u = \frac{x}{\sqrt{1+x^2}}$

i.e. $\sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}$

SOLUTION

(b) $\tan(\sin^{-1} x)$

$\boxed{\text{Assuming } \sin^{-1} x = u}$

$\sin u = x$, $[-\frac{\pi}{2} \leq u \leq \frac{\pi}{2}]$

From First Pythagorean identity-

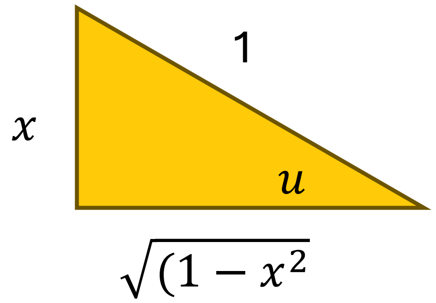
$\cos u = + \sqrt{1 - x^2}$

We know that,

$\tan u = \frac{\sin u}{\cos u}$

i.e. $\tan u = \frac{x}{\sqrt{1-x^2}}$

i.e. $\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}$



SOLUTION